

Set 12: Website Traffic Analysis

DSA – Data Handling and Visualization | Lab Experiment

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Aim

To analyse daily website traffic using a line chart of page views, a Top-N bar chart of click-through rate and a labelled table, and to combine them in an interactive Tableau dashboard.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Date	Page Views	Click-through Rate
2023-01-01	1,500	2.3%
2023-01-02	1,600	2.7%
2023-01-03	1,400	2.0%

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
traffic <- data.frame(
  Date      = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Page_Views = c(1500, 1600, 1400),
  CTR       = c(2.3, 2.7, 2.0)          # click-through rate in %
)
traffic$Clicks <- round(traffic$Page_Views * traffic$CTR / 100)
write.csv(traffic, "Website_Traffic.csv", row.names = FALSE) # for Tableau
```

Q1. Line Chart – Daily Page Views Trend

Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

R Program:

```
# Q1. Line chart - daily page views trend
print(traffic[, c("Date", "Page_Views")], row.names = FALSE)
cat("Day-over-day change:", diff(traffic$Page_Views), "\n")
cat("Average page views :", round(mean(traffic$Page_Views), 1), "\n")

p_line <- ggplot(traffic, aes(x = Date, y = Page_Views)) +
  geom_line(colour = "#0072B2", linewidth = 1.2) +
  geom_point(size = 4, colour = "#0072B2") +
  geom_text(aes(label = format(Page_Views, big.mark = ",")), vjust = -1.2, size = 4) +
  scale_x_date(date_labels = "%d %b %Y", breaks = traffic$Date,
    expand = expansion(mult = 0.12)) +
  scale_y_continuous(limits = c(1300, 1700), labels = scales::label_comma()) +
  labs(title = "Daily Page Views Trend", x = "Date", y = "Page Views") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_line)
ggsave("s12_q1_line.png", p_line, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
      Date Page_Views
2023-01-01      1500
2023-01-02      1600
2023-01-03      1400
Day-over-day change: 100 -200
Average page views : 1500
```

Output:

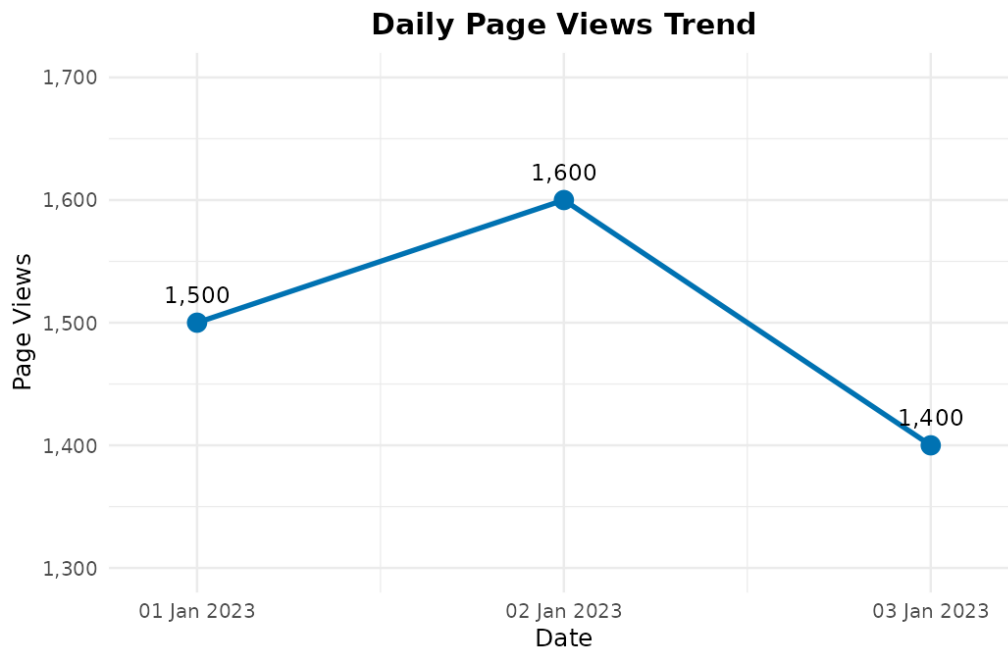


Figure 1: Daily page views

Inference:

Page views rose by 100 from 1,500 on 1 Jan to a peak of 1,600 on 2 Jan, then fell by 200 to 1,400 on 3 Jan. The average is 1,500 per day. Traffic peaked mid-period and the most recent day was the lowest.

Q2. Bar Chart – Top N Days by Click-through Rate

Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

R Program:

```
# Q2. Bar chart - top N days by click-through rate
N <- 2
top_n_days <- head(traffic[order(-traffic$CTR), ], N)
cat("Top", N, "days by CTR:\n"); print(top_n_days, row.names = FALSE)

top_n_days$Day <- factor(format(top_n_days$Date, "%d %b %Y"),
  levels = format(top_n_days$Date, "%d %b %Y"))
p_bar <- ggplot(top_n_days, aes(x = Day, y = CTR)) +
  geom_col(fill = "#009E73", width = 0.5) +
  geom_text(aes(label = paste0(CTR, "%")), vjust = -0.5, size = 4.5) +
  scale_y_continuous(limits = c(0, 3.2), labels = function(x) paste0(x, "%")) +
  labs(title = paste("Top", N, "Days with Highest Click-through Rate"),
    x = "Date", y = "Click-through Rate (%)") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_bar)
ggsave("s12_q2_topN.png", p_bar, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
Top 2 days by CTR:
  Date Page_Views CTR Clicks
2023-01-02      1600  2.7    43
2023-01-01      1500  2.3    34
```

Output:

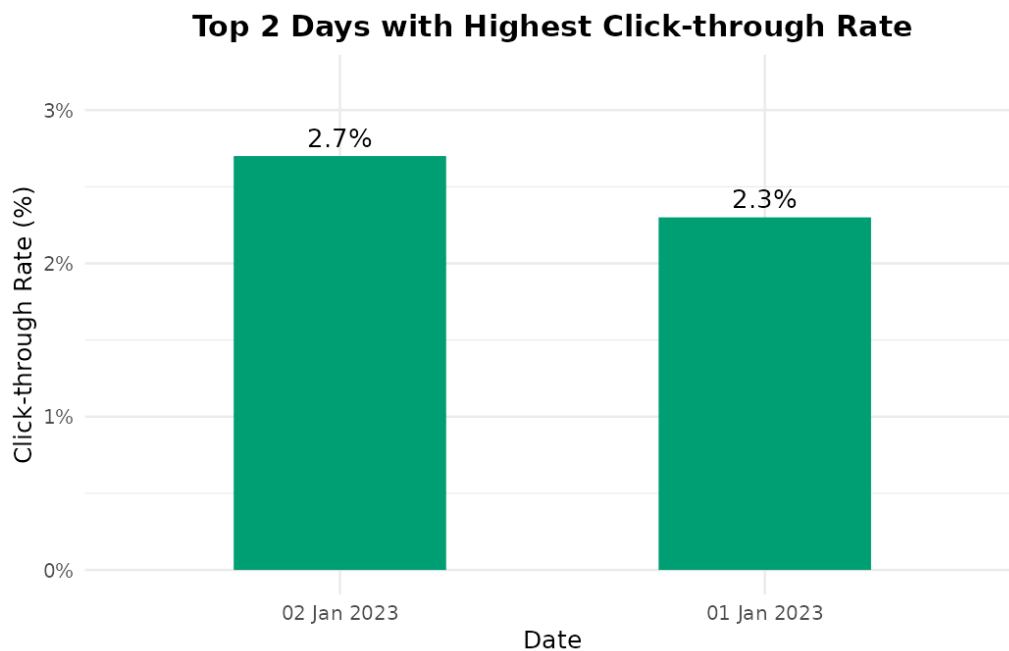


Figure 2: Top 2 days by click-through rate (N = 2)

Inference:

With N = 2, the top days are 2 Jan (2.7%) and 1 Jan (2.3%). The day with the most page views (2 Jan) also had the best CTR, about 43 clicks. Change N in the program to show more or fewer days.

Q3. Table – Website Traffic Data for Each Day

Build a table to show the website traffic data for each day. Label the table elements.

R Program:

```
# Q3. Labelled table of website traffic
tbl <- traffic
tbl$Date <- format(tbl$Date, "%Y-%m-%d")
tbl$page_views <- format(tbl$page_views, big.mark = ",")
tbl$CTR <- sprintf("%.1f%", tbl$CTR)
names(tbl) <- c("Date", "Page Views", "Click-through Rate", "Clicks (derived)")
print(tbl, row.names = FALSE)
tg <- tableGrob(tbl, rows = NULL, theme = ttheme_default(
  core = list(fg_params = list(fontsize = 11)),
  colhead = list(fg_params = list(fontsize = 12, fontface = "bold", col = "white"),
    bg_params = list(fill = "#1F3864"))))
title <- grid::textGrob("Website Traffic Data by Day",
  gp = grid::gpar(fontsize = 14, fontface = "bold"))
png("s12_q3_table.png", width = 1000, height = 330, res = 150)
grid.arrange(title, tg, ncol = 1, heights = c(0.25, 1))
invisible(dev.off())
```

Console Output:

Date	Page Views	Click-through Rate	Clicks (derived)
2023-01-01	1,500	2.3%	34
2023-01-02	1,600	2.7%	43
2023-01-03	1,400	2.0%	28

Output:

Website Traffic Data by Day

Date	Page Views	Click-through Rate	Clicks (derived)
2023-01-01	1,500	2.3%	34
2023-01-02	1,600	2.7%	43
2023-01-03	1,400	2.0%	28

Figure 3: Labelled website traffic table

Inference:

The table adds a derived **Clicks** column (Page Views × CTR): 34, 43 and 28 clicks. 3 Jan was the weakest day on views, CTR and clicks.

Q4. Tableau Dashboard – Interactive Exploration

Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of website traffic data.

R Program:

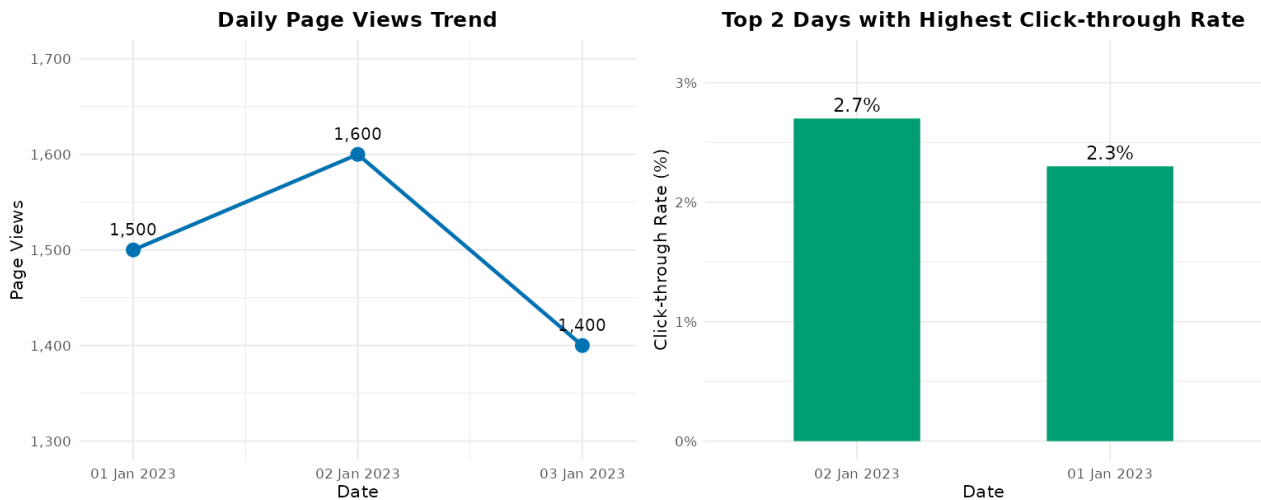
```
# Q4. Dashboard layout preview (Tableau dashboard built from Website_Traffic.csv)
library(patchwork)
dash <- (p_line | p_bar) / wrap_elements(full = tg) +
  plot_layout(heights = c(1.4, 0.6)) +
  plot_annotation(title = "Website Traffic Dashboard",
                  theme = theme(plot.title = element_text(face = "bold", size = 16)))
ggsave("s12_dashboard.png", dash, width = 12, height = 7, dpi = 130, bg = "white")
```

Procedure in Tableau:

- 1 Open Tableau → **Connect** → **Text file** → **Website_Traffic.csv**. Make sure Date is typed as **Date**. CTR is stored as a number (e.g. 2.3), so create a calculated field **CTR % = [CTR] / 100** and format it as Percentage (1 decimal).
- 2 **Sheet 1 – "Page Views Trend"**: drag Date to Columns and set it to **Exact Date / Day** (continuous). Drag SUM(Page Views) to Rows and set the Marks type to Line with markers and labels. Title: "Daily Page Views Trend".
- 3 **Sheet 2 – "Top N CTR Days"**: create an integer parameter **Top N** (range 1–3, default 2) → right-click → Show Parameter. Drag Date (discrete Day) to Columns and AVG(CTR %) to Rows. Drag Date to Filters → **Top** tab → By field: **Top [Top N] by AVG(CTR %)**. Sort descending and show labels.
- 4 **Sheet 3 – "Traffic Table"**: drag Date (discrete) to Rows and Measure Names to Columns. Put Measure Values (Page Views, CTR %, Clicks) on Text, and rename the headers.
- 5 From the menu choose **Dashboard** → **New Dashboard**. Set **Size** → **Automatic** (or Fixed 1200 × 800).
- 6 Drag the sheets onto the canvas: Page Views Trend (top-left), Top N CTR Days (top-right), Traffic Table (bottom); also place the Top N parameter control. Add a **Text** object at the top for the dashboard title.
- 7 On each chart, click the sheet's drop-down and select **Use as Filter**. Clicking a bar or slice then filters the other views.
- 8 Go to **Dashboard** → **Actions** → **Add Action** → **Highlight** (Run on: Hover) so related marks highlight across all views.
- 9 Right-click a filter → **Apply to Worksheets** → **All Using This Data Source** (Website_Traffic) so one filter controls every view.
- 10 Save as a packaged workbook (**File** → **Save As** → **.twbx**) or publish to Tableau Public.

Dashboard Layout:

Website Traffic Dashboard



Reference layout of the dashboard (rendered in R). The interactive version is built in Tableau using the procedure above.

Inference:

The Top N parameter lets users change N interactively. Selecting a date on the line chart highlights the same date in the CTR bar chart and filters the table to that day.

Result

The website traffic data was visualised with a line chart, a Top-N bar chart and a labelled table in R, and combined into an interactive Tableau dashboard with a Top-N parameter. Page views peaked on 2 Jan 2023 (1,600), which was also the day with the highest click-through rate (2.7%).